PREDICTION IS NOT THE PRODUCT.

RTO Shield converts COD order risk prediction into **financially optimal intervention decisions** for Indian e-commerce merchants — a measured, frozen, independently reproduced decision engine.

THE PROBLEM

Cash-on-delivery dominates Indian e-commerce, and **28.27% of test COD orders** came back as Return-to-Origin — each costing **₹150** in reverse logistics. Low-value risky orders are economically unsalvageable; high-value ones deserve a frictionless path.

THE SYSTEM

A calibrated LightGBM scores P(RTO) from **15 order-time features** — 13 merchant inputs plus 2 fields resolved from a train-only pincode lookup. An expected-loss engine prices every intervention per order — `EL(action) = friction + P_rto_after × ₹150 − P_success × margin` — and routes each order to the argmin: **ALLOW**, **VERIFY** (₹2 friction, −30% RTO), or **DEPOSIT** (−80% RTO). A fourth option, PREPAID_ONLY, is proven mathematically dominated and never selected.

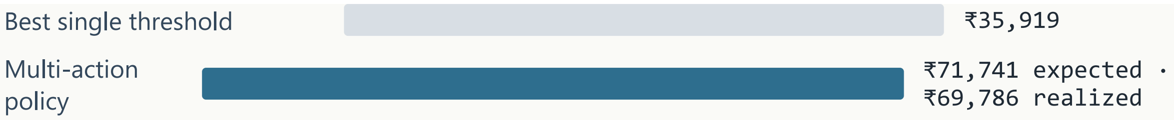
THE PROOF

On a strictly held-out test window (7,174 COD orders), the multi-action policy earned **₹71,741 expected savings** and **₹69,786 realized** against actual labels — −2.7% vs forecast, inside Monte Carlo noise. The uncalibrated router's forecast collapsed **−15.3%** against the same labels: calibration bought the only forecast that held up.

13.1%	2.0×	94.7%	100%
COD PORTFOLIO PROFIT UPLIFT	SAVINGS VS BEST SINGLE THRESHOLD	OF MEASURED BAYES CEILING	P(SAVINGS > 0) 5,000 DRAWS

SAVINGS VS ALWAYS-ALLOW — HELD-OUT TEST WINDOW

Always allow (baseline) ₹0



60/60 TESTS PASSING · 11/11 PRE-REGISTERED CHECKS · COLD-CLONE REPRODUCED

LIVE DEMO: rto-shield-nlthpydtndpkupyfgfl3yy.streamlit.app · github.com/mahirbhat70-eng/rto-shield · MIT